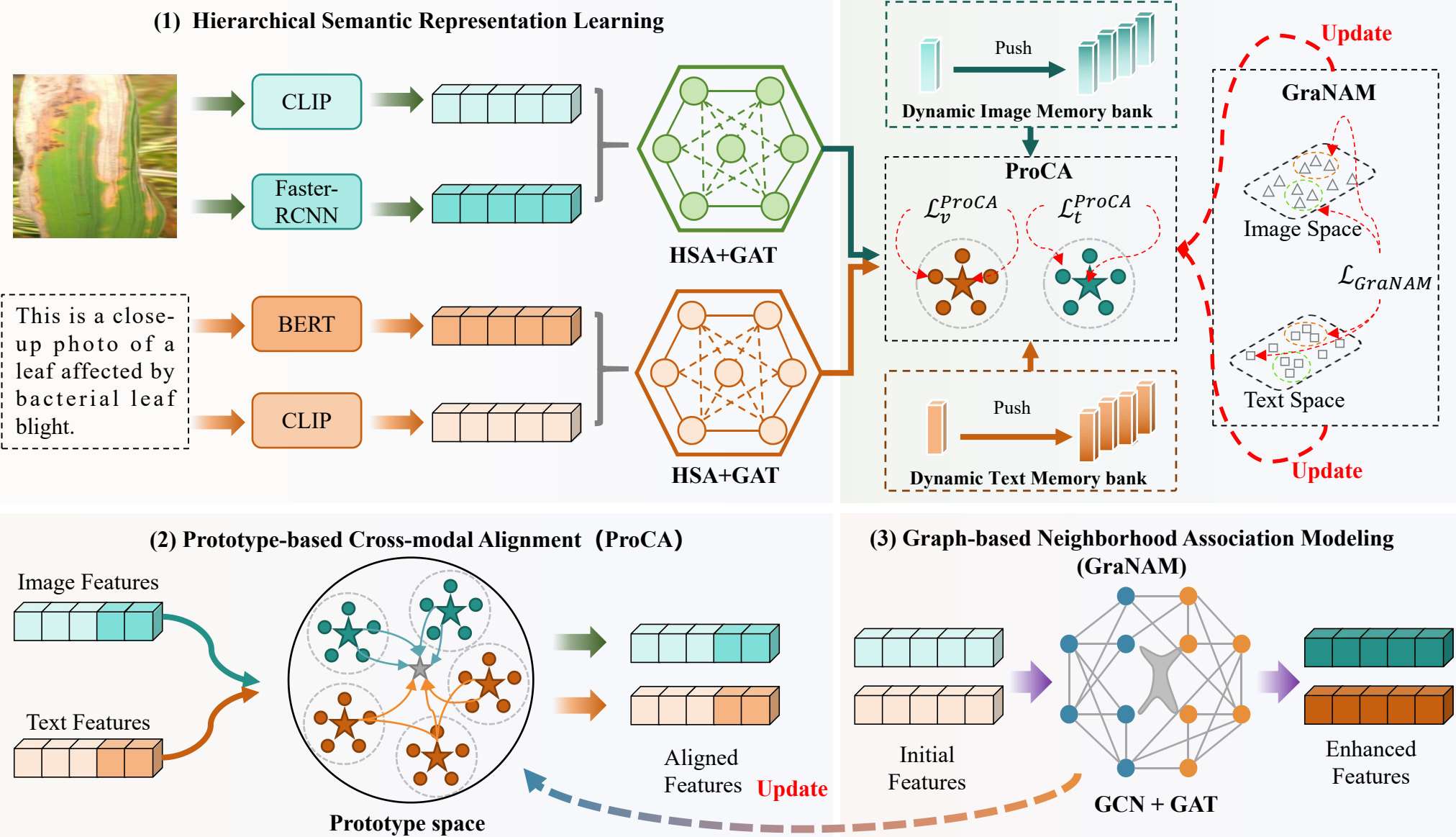

(1) Hierarchical Semantic Representation Learning
CLIP
Faster-RCNN
HSA+GAT
This is a close-up photo of a leaf affected by bacterial leaf blight.
BERT
CLIP
HSA+GAT
Push
Dynamic Image Memory bank
ProCA
$\mathcal{L}_v^{ProCA}$
$\mathcal{L}_t^{ProCA}$
Push
Dynamic Text Memory bank
Update
GraNAM
Image Space
$\mathcal{L}_{GraNAM}$
Text Space
Update
(2) Prototype-based Cross-modal Alignment (ProCA)
Image Features
Text Features
Prototype space
Aligned Features
Update
(3) Graph-based Neighborhood Association Modeling (GraNAM)
Initial Features
GCN + GAT
Enhanced Features